

Elisha Erzoah

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EDUCATION

- **George Mason University** Fairfax, VA, USA
Doctor of Philosophy in Mathematics Aug. 2024 – Present
- **African Institutes for Mathematical Sciences(AMIS)-Ghana** Accra, Ghana
Masters of Science in Mathematical Science Nov. 2022 – Aug. 2023
- **University of Cape Coast** Cape Coast, Ghana
Bachelor of Science in Mathematics with Economics Aug. 2017 – Sept. 2021

RESEARCH INTERESTS

Mathematical and Computational Biology, Mathematical Oncology, Network Biology, Applied Mathematics, Data Science, Dynamical Systems, Data Science, Machine Learning

EXPERIENCE

- **Department of Mathematics, George Mason University** Fairfax, VA, USA
Graduate Research Assistant Aug. 2025 - Present
 - **Responsibilities:**
 - * Conducting research in mathematical biology, developing data-driven models to investigate the dynamics and adaptation mechanisms of the MAPK signaling pathway.
 - * Presenting findings at lab meetings and academic conferences
- **Department of Mathematics, George Mason University** Fairfax, VA, USA
Graduate Teaching Assistant Aug. 2024 - May 2025
 - **Responsibilities:** In the role of Teaching Assistant, I was tasked with:
 - * Led structured recitation sessions designed to strengthen conceptual understanding and analytical skills.
 - * Held regular office hours to provide individualized academic support and mentoring to students.
 - * Graded quizzes and assignments, providing constructive feedback to support student learning.
 - **Courses Taught:** Calculus I & Abstract Algebra I .
- **Industry Immersion Africa (Ghana 1000)** Accra, Ghana
Tutor Sept. 2023 - Nov. 2023
 - **About the Ghana 1000 program:** The Ghana 1000 program is an annual initiative launched as part of the Industry Immersion Africa. It spans three months and aims to provide comprehensive training to 1000 STEM graduates, equipping them with the skills needed to thrive in the competitive business environment.
 - **Responsibilities:** In the role of Tutor, I was tasked with:
 - * Delivered engaging course material to explain complex concepts and ensure student understanding.
 - * Graded assignments, quizzes, and exams, providing detailed and constructive feedback to support student growth.
 - * Responded promptly to student questions during office hours and via email, creating an accessible and encouraging learning environment.
 - **Courses Tutored:** Introduction to Power BI, Advanced Microsoft Excel for Data Analytic, Data and Decisions, Design Thinking
 - **Tools:** Power BI and Microsoft Excel
- **Department of Mathematics, University of Cape Coast** Cape Coast, Ghana
Teaching Assistant Oct. 2021 - Sept. 2022
 - **Responsibilities:** In the role of Teaching Assistant, I was tasked with:
 - * Organizing tutorials for students.
 - * Invigilating quizzes and semester examinations.
 - * Marking and grading examination papers.
 - **Courses Taught:** Abstract Algebra I & II, Complex Numbers and Matrix Algebra.

RESEARCH

- **Industrial Immersion Program (IIP)** Jun. 2026 – Present
Graduate Researcher
 - **Machine Learning-Based Parameter Estimation Using Physics-Informed Neural Networks (PINNs) and Kolmogorov-Arnold Networks (KANs)**
Advisor: Dr. Elie Alhajjar, RAND Corporation, Arlington, VA
- **George Mason University** Aug. 2025 – May 2026
Graduate Researcher
 - **Data-Driven Modeling of Dynamics and Adaptation in Mitogen Activated Protein Kinase(MAPK) Pathway**
Advisor: Prof. Maria Emelianenko, Department of Mathematics, GMU
- **African Institute for Mathematical Science (AIMS) - Ghana** May 2023 – Aug. 2023
Graduate Researcher
 - **Oscillatory Dynamics In Cell Signaling**
Advisor: Prof. Nick Monk, Department of Mathematics, University of Sheffield & AMIS Ghana

RESEARCH FUNDING

- **Industrial Immersion Program (IIP)** Jun. 2026 – Present
Department of Mathematical Sciences, GMU, Fairfax, VA
- **Graduate Research Assistant** Aug. 2025 – May 2026
Office of the Provost, GMU, Fairfax, VA
- **Graduate Teaching Assistant** Aug. 2024 – May 2025
Department of Mathematical Sciences, GMU, Fairfax, VA

TRAVELING GRANT

- ICM Travel Grant to attend ICM 2026 Satellite Conference on Industrial and Computational Mathematics held at Morgan State University Jun. 1-5, 2026
- Graduate Student Travel Grant (GSTG) to attend Joint Mathematics Meeting (JMM) 2026 Jan. 4-7, 2026

ORAL PRESENTATION

- **Data-Driven Integrated Framework for Modeling Protein Interaction Networks** June 3, 2026
ICM 2026 Satellite Conference on Industrial and Computational Mathematics, MSU, Baltimore, MD
- **Data-Driven Modeling of Dynamics in MAPK Pathway** Nov. 9, 2025
Symposium on BEER, George Mason University, Fairfax, VA

MENTORING

- Mason Math Odyssey Enrichment Camp organized at George Mason University July 7-18, 2025

FELLOWSHIP PARTICIPATION

- **Foundation Name:** MasterCard Foundation (Scholars Program)
- **Foundation Overview:** The Mastercard Foundation Scholars Program envisions a network of young leaders and institutions driving inclusive socio-economic change in Africa. The goal of the foundation is to bridge the educational gap and provide quality education to underserved communities across Africa, fostering equity and opportunity.
- As a member of MasterCard foundation(MasterCard Scholar), I actively engaged in a series give-back to community initiatives organised by the foundation.
 - Visitation to orphanage homes.
 - Participation in community clean-up exercises.
 - Engagement in teaching activities across elementary, junior and senior high school levels.
 - Collaborated in organizing health screening activities in various less endowed communities.
 - Guidance and mentorship for students.

TECHNICAL SKILLS

- **Programming Languages:** Python and Matlab
- **Data Analysis & Visualization:** Microsoft Excel (Advanced Formulas, PivotTables) and Power BI
- **Scientific Typesetting & Documentation:** Latex, Microsoft Office Suite (Word, PowerPoint)